

DATABASES – seminar 6

The difference between 'YYYY' and 'RRRR' in Oracle SQL

SYSDATE YEAR	SPECIFIED DATE	STORED WITH RR	STORED WITH YY
1995	07-nov-95	1995	1995
1995	07-nov-12	2012	1912
2020	07-nov-95	1995	2095
2020	07-nov-12	2012	2012
2063	07-nov-95	2095	2095
2063	07-nov-12	2112	2012

select to_char(to_date('07-nov-95','dd-mon-rr'),'rrrr') from dual;

select to_char(to_date('07-nov-95','dd-mon-yy'),'rrrr') from dual;

select to_char(to_date('07-nov-12','dd-mon-rr'),'rrrr') from dual;

select to_char(to_date('07-nov-12','dd-mon-yy'),'rrrr') from dual;

Oracle NUMBER data type

The **precision** is the **number** of digits in a **number**. It ranges from 1 to 38. The **scale** is the **number** of digits to the right of the decimal point in a **number**. It ranges from -84 to 127.

Input Data	Specified as	Stored as
1295.46	NUMBER	1295.46
1295.46	NUMBER(*,1)	1295.5
1295.46	NUMBER(6)	1295
1295.46	NUMBER(6,2)	1295.46
1295.46	NUMBER(6,1)	1295.5
1295.46	NUMBER(3)	-
1295.46	NUMBER(6,-2)	1300
.00000046	NUMBER(2,7)	0.0000005

create table ex(a number(2,7));

insert into ex values (.00000046);

insert into ex values (.0000099);

insert into ex values (.00000999);

insert into ex values (.00000991);

The following example uses the `bonuses` table with a default bonus of 100. It then inserts into the `bonuses` table all employees who made sales, based on the `sales_rep_id` column of the `orders` table. Finally, the human resources manager decides that employees with a salary of \$8000 or less should receive a bonus. Those who have not made sales get a bonus of 1% of their salary. Those who already made sales get an increase in their bonus equal to 1% of their salary. The `MERGE` statement implements these changes in one step:

```
CREATE TABLE bonuses (employee_id NUMBER, bonus NUMBER DEFAULT 100);
```

```
INSERT INTO bonuses(employee_id)
  (SELECT e.employee_id FROM employees e, orders o
   WHERE e.employee_id = o.sales_rep_id
   GROUP BY e.employee_id);
```

```
SELECT * FROM bonuses ORDER BY employee_id;
```

```
MERGE INTO bonuses D
  USING (SELECT employee_id, salary, department_id FROM employees
   WHERE department_id = 80) S
  ON (D.employee_id = S.employee_id)
  WHEN MATCHED THEN UPDATE SET D.bonus = D.bonus + S.salary*.01
  DELETE WHERE (S.salary > 8000)
  WHEN NOT MATCHED THEN INSERT (D.employee_id, D.bonus)
  VALUES (S.employee_id, S.salary*.01)
  WHERE (S.salary <= 8000);
```

```
SELECT * FROM bonuses ORDER BY employee_id;
```